

Hidden Treasure

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

On a random afternoon, Bob and Alice's grandfather decided to give them what he claims to be his most valuable possession: the map to his hidden treasure.

Their city is represented by a 2D grid that goes from $(-100, -100)$ to $(100, 100)$, and the map, instead of functioning like a normal map, contains n "anchor" points. Each point is represented by x_i , y_i and d_i : the position of the anchor point and its Manhattan distance from the treasure.

Recall that the Manhattan distance between two points (x_1, y_1) and (x_2, y_2) is calculated as:

$$|x_1 - x_2| + |y_1 - y_2|$$

However, they think their grandfather is lying, so they ask for your help to either find the treasure, or report that it doesn't exist.

Input

The first line contains an integer n ($1 \leq n \leq 1000$) — the number of anchor points in the map.

Each of the following n lines contains three integers x_i , y_i and d_i ($-100 \leq x_i, y_i \leq 100$; $0 \leq d_i \leq 400$) — the coordinates of the anchor point and the Manhattan distance to the treasure, respectively.

Output

Output two space-separated integers X and Y ($-100 \leq X, Y \leq 100$) — the coordinates of the treasure. If there is no coordinate that matches the description in the map, report that the treasure doesn't exist by outputting -1 .

If there are multiple coordinates that match the description in the map, you can output any of them.

Examples

standard input	standard output
3 0 0 7 3 0 4 6 8 7	3 4
2 0 5 5 -5 0 5	0 0
2 0 5 5 -10 0 1	-1

Note

- **Example 1:** Anchors at $(0, 0, 7)$, $(3, 0, 4)$, and $(6, 8, 7)$.
 - If we test the point $(3, 4)$, its distance to $(0, 0)$ is $|3 - 0| + |4 - 0| = 7$.
 - Its distance to $(3, 0)$ is $|3 - 3| + |4 - 0| = 4$.
 - Its distance to $(6, 8)$ is $|6 - 3| + |8 - 4| = 3 + 4 = 7$.

- It perfectly matches the required distance for all anchors. Output: 3 4.
- **Example 3:** Anchors at $(0, 5, 5)$ and $(-10, 0, 1)$.
 - The first anchor tells us the treasure must be exactly 5 steps away from $(0, 5)$.
 - The second anchor requires the treasure to be 1 step away from $(-10, 0)$.
 - There is no coordinate on the grid that can simultaneously be that close to $(-10, 0)$ while still reaching $(0, 5)$. It is mathematically impossible. Output: -1.